Skony — Track 2A OpenParlData: Technical Report

Repository: <https://github.com/ahmarkhhan/sfedly> (built from the HackApertus project template). Build and completion evidence: BUILD_STATUS. Shared affair records and evaluation remain planned. Response caching, batch checkpoints, saved-result import and gold-export safeguards are implemented. Docker packaging and a synthetic practice replay exist; actual container execution is still unverified. Pipeline evidence.

Status: extraction and local review implementation, 16 October 2026. One real-document pilot is complete; dataset-level extraction accuracy has not been measured. Measured run evidence: RESULTS.

Problem and contribution

Investigate which parliamentary affair fields can share a common representation across bodies and languages. Preserve original wording, explicitly abstain when evidence cannot be located, and record where the shared schema cannot represent the source faithfully.

Implemented first slice

PDF → page text and digest → Apertus schema-constrained prompt → strict JSON validation → evidence lookup → review-required JSON. The output has eight raw fields. Normalisation and controlled vocabularies are planned, rather than inferred and silently treated as source text.

Each non-null result includes a document ID, page number, original quote and character offsets into the unchanged extracted page text. `source_located` means a string was found; it does not establish correct field assignment or faithfulness of question/answer boundaries. All such results require review.

Validation so far

131 engineering tests passed on local Python 3.11 (RESULTS). The client was exercised with mocked HTTP transports and the real CSCS service. Tests cover fabricated quotes/values, wrong page attribution, repeated quotes, missing evidence, textless/oversize input, malformed schemas, truncated/mismatched endpoint responses, line-break span recovery and obviously corrupt submitter OCR. Review tests also cover persistence, stale revisions, fabricated correction rejection, explicit absence, visual transcription, audit snapshots, model reruns preserving human decisions, reviewed-only export and local route/secret isolation.

New execution checks cover replay blocking live calls, synthetic-fixture hashes, restart persistence and mapped localhost Host/Origin checks. A clean public-source copy starts without private data/keys and serves the PDF/image and reviewed export. The bundled practice data are authored examples, not model output or evaluation gold.

The batch/cache checks cover full effective-request identity, corrupt-cache rejection, schema-invalid response replay, actual call accounting, preserved failed downloads, unchanged manifests, interrupted resume and idempotent workbench import. Gold-export checks exclude synthetic/QA records and prevent stale labels from a previous export directory entering a new manifest.

An approved real-data engineering proof processed one DE Nidau PDF and one FR Genève PDF using Apertus 1.5 70B at CSCS. The first run made two calls, reporting 3,953 total tokens and 13.594/9.406 seconds. Offline replay, with no credential and an HTTP-rejecting transport, used two cached responses and produced identical verified results with zero model calls. One failed-fetch row remained in both ledgers. This demonstrates pipeline reuse, not semantic accuracy or multilingual benchmark coverage. Both results were imported without altering existing review decisions. The store has 12 decisions across four documents, one mechanically complete; it is not yet a frozen, competence-checked evaluation gold set.

On one scanned Nidau postulate with an existing OCR layer, the 8B pilot first copied the schema and then returned inconsistent null/evidence combinations. Both were rejected. The 70B pilot returned valid fields. Replaying that recorded response with verifier `evidence-v0.2` located the title, type, printed date and request; it abstained on the garbled submitter and absent answer/decision fields. The 70B call took 7.234 seconds and used 1,725 tokens. These are single-call observations, not a benchmark or a gold-set result.

Planned evaluation

Use separate development and held-out human-reviewed documents spanning German and French and multiple parliamentary bodies. Compare a fair simple baseline, raw Apertus extraction and extraction with evidence checks. Report field precision/recall, abstention/coverage, source-location accuracy, question/answer pairing, latency and actual

token usage. Scans and minority layouts must remain visible in the report rather than being filtered out of the denominator.

Current limitations

The local workbench displays PDF page images, source text and highlighted spans. Reviewers can confirm, correct, record absence or undo. Decisions are separate from model suggestions in SQLite, with before/after audit history. Visual transcriptions are explicitly labelled and carry no claimed text offsets. Exports contain reviewed fields only, plus a list of remaining unreviewed fields. Reviewer identity is a self-entered label; multi-user authentication is not implemented.

No OCR, multi-page field assembly, controlled vocabulary mapping, cross-body query interface or gold-set evaluation is implemented yet. The Docker runner is packaged but has not been built/run in this environment. Whitespace-only matching recovers some layout differences but preserves wording, punctuation and original offsets. Other OCR/layout differences still require review; the OCR marker heuristic does not detect all corrupt names. Copied text can still be assigned to the wrong semantic field. Model identity must be confirmed against the provider's documented served-model mapping.

Deployment and licensing

The application uses a configurable Apertus endpoint. A Swiss endpoint and self-hosted endpoint are deployment targets; air-gapped operation has not been tested.

Component	Licence	Note
Application code (this repository)	Apache-2.0	per event terms
Documentation and this report	CC-BY-4.0	per event terms
Bundled synthetic fixture (`data/sample/`)	CDLA-Permissive-2.0	authored; fictional people and parliament
Runtime dependencies: pydantic, httpx, pypdf (pinned in `requirements.txt`)	MIT / BSD-3-Clause	check each package `LICENSE` before redistribution
Apertus v1.5 via the CSCS endpoint (development time)	Apache-2.0 (model)	not redistributed
OpenParlData source PDFs (private corpus)	source-specific terms	not redistributed; a source manifest must state item-level licences before any redistribution or before a dataset licence is applied to our own annotations
Container base (`python:3.11-slim`, Poppler)	Debian package terms	runner image only